

S.No	Author Name, Journal, Publication Year	Research/ Methodology	Performance Metrics	Advantages	Disadvantages
1.	Lili Li, Shujuan Zhang & Bin Wang, IEEE Access, 2021	Review of deep learning methods for plant disease detection and classification, mainly focusing on image-based approaches such as CNNs. Dataset: PlantVillage Dataset — publicly available crop/plant disease image dataset.	Reviews accuracy and classification performance reported by existing deep-learning studies.	Automated detection, high classification capability, reduces manual effort, and supports early disease identification.	Requires large and diverse datasets; performance can be affected by image quality and real-field conditions.
2.	A. Dolatabadian, T. X. Neik, M. F. Danilevicz, S. R. Upadhyaya, J. Batley & D. Edwards, Plant Pathology, 2025	Review of image-based crop disease detection using ML/DL, CNNs, imaging sensors and contextual data. Dataset: PlantVillage, PlantDoc, LeafSnap, AI Challenger Plant Disease Recognition, and other self-collected datasets.	Reviews accuracy and other performance measures reported by different crop-disease detection studies.	Fast and automated detection; ML can identify visual disease patterns; supports different crops, imaging platforms and real-time monitoring.	Performance can decrease in real-field conditions due to varying lighting, background, environmental conditions and limited contextual data.
3.	Munir Majdalawieh, Carla Martins, Mohammed Radi, Maher Alaraj & Shafaq Khan, Smart Agricultural Technology, 2025	Systematic review of ML/AI methods for crop disease detection, including CNNs, Vision Transformers (ViTs), datasets and evaluation methods.	Reviews accuracy and other performance metrics reported by existing crop-disease detection models.	CNNs are cost-effective and widely used; ViTs can achieve higher accuracy; supports precision agriculture.	ViTs require high computational resources; dataset bias, limited real-time applicability, and high infrastructure cost can affect practical adoption.

		Dataset: Multiple existing crop-disease image datasets reported in previous studies, including PlantVillage and other crop-specific/field datasets.			
4.	Ammar Oad, Syed Shoaib Abbas, Amna Zafar, Beenish Ayesha Akram, Feng Dong, Mir Sajjad Hussain, Mune Uddin, IEEE Access, 2024	Ensemble learning combined with Explainable AI (XAI) for plant leaf disease detection and classification. Dataset: PlantVillage Dataset	Accuracy, precision, recall, and F1-score are used to evaluate the model.	Improves disease classification; ensemble learning increases robustness; XAI makes predictions more understandable.	Higher computational complexity; requires suitable training data and computational resources.
5.	Anuja Bhargava, Aasheesh Shukla, Om Prakash Goswami, Mohammed H. Alsharif, Peerapong Uthansakul & Monthippa Uthansakul, IEEE Access, 2024	Review of computer vision and AI techniques for plant leaf disease detection, classification, and diagnosis. Dataset: PlantVillage + Kaggle plant disease datasets + self-collected datasets	Reviews accuracy, precision, recall, F1-score and other metrics reported by existing studies.	Automated detection, reduces manual effort, supports early diagnosis, and covers various AI/computer-vision approaches.	Performance depends on dataset quality; variations in real-field images, lighting, background, and disease appearance can affect detection.
6.	Abdullah Al Mamun, David Ahmed-Aristizabal, Miaohua Zhang, Md Ismail Hossen, Zeeshan Hayder & Mohammad Awrangjeb, IEEE Access, 2024	Systematic review of self-supervised learning (SSL) techniques for plant disease detection, focusing on learning useful features with limited labelled data.	Reviews accuracy, precision, recall, F1-score and other metrics reported by existing SSL-based studies.	Reduces dependence on labelled datasets; useful when labelled agricultural images are limited; can learn meaningful image features.	Training can be complex; performance depends on the quality and diversity of images; real-field variations can still affect detection.

		Datasets: PlantVillage, Citrus Disease, BRACOL, PlantDoc, IP102, Citrus Pest Benchmark, Paddy Doctor, and Corn Disease and Severity Dataset.			
7.	Srinu Banothu, Karnam Madhavi, K. M. V. Madan Kumar, Ramesh Gajula, Ch. Mallikarjuna Rao, Saurav Dixit & Abhishek Chhetri, Cogent Engineering, 2024	DenseNet is used for plant disease identification, followed by pesticide recommendation based on the detected disease. Dataset: PlantVillage Dataset — New Plant Village Dataset (Test Set)	Accuracy, precision, recall and F1-score for disease identification.	Combines disease identification and pesticide recommendation; automated; reduces manual diagnosis effort.	Requires quality training images; performance may be affected by real-field conditions and different disease symptoms.
8.	Natasha Nigar, Hafiz Muhammad Faisal, Muhammad Umer, Olukayode Oki, and Jose Manappattukunnel Lukose, IEEE Access, 2024	Deep-learning- based prediction model for plant disease classification combined with Explainable AI (XAI). Dataset : New Plant Disease Dataset / PlantVillage- based dataset	Accuracy, precision, recall and F1-score are used to evaluate classification performance.	Improves disease classification; XAI provides understandable explanations of model predictions; supports automated diagnosis.	Requires sufficient quality training data; deep-learning models can require high computational resources; real- field variations may affect performance.
9.	Ayesha Khaliq, Atif Khan, Salman Jan, Muhammad Umair, Asad Gulshair, Ahmad Ali & Usman Ali Shah, IEEE Access, 2025	AI-based smart agriculture using TabNet for fertilizer recommendation, SwiFT for crop recommendation, TTL for irrigation, and XAI for	Fertilizer recommendation: 99.3% accuracy; Crop recommendation: 98.75% accuracy; Soil analysis: $R^2 =$ 98.7%, RMSE = 1.51; Irrigation:	High prediction accuracy; provides personalized fertilizer recommendations; XAI improves explainability; uses real-time soil information.	Requires soil/environmental data and sensors: system is more complex: performance/ generalization may vary across

		explaining predictions. Dataset: PlantVillage	99.13% accuracy.		different geographical regions.
10.	Christine Musanase, Anthony Vodacek, Damien Hanyurwimfura, Alfred Uwitonze & Innocent Kabandana, Agriculture, 2023	Neural Network for crop recommendation And a rule-based fertilizer recommendation system using Soil nutrients such as N, P, K and soil PH Dataset: Crop Recommendation Dataset — Kaggle	97% accuracy for the proposed prediction model.	Personalized fertilizer recommendations; combines ML with agricultural domain knowledge; supports precision farming and IoT-based data collection.	Deceloped specifically for Rwanda: needs additional environmental/ geographical factors such as rainfall, temperature and humidity for broader applicability,
11.	C. J. Xan et al., IEEE Access, 2024	MobileNet V1/V2, FD-MobileNet, ResNet and SqueezeNet optimized for ARM-M microcontroller deployment Dataset: 5,932 rice-leaf images	FD-MobileNet: 98.44% accuracy	Lightweight and suitable for real-time edge deployment	Limited to rice diseases and resource-constrained hardware
12.	A. Prommakhot et al., IEEE Access, 2025	Hybrid CNN + BiLSTM + Transformer sequential-learning architecture Dataset: PlantVillage, 38 plant-disease classes	97.88%accuracy, 97.93%precision, 97.62% recall	Combines CNN feature extraction with sequential/Transformer learning	More computationally intensive than conventional CNNs
13.	P. Chaisiriprasert, K. Chuiad, IEEE Access, 2025	Lightweight Color-Aware Transformer with hierarchical attention Dataset: Plant Disease Classification Merged Dataset	75% accuracy, 0.81 mAP; 3.33G FLOPs vs 17.58G for ViT	Lightweight Transformer; suitable for resource-constrained deployment	Accuracy is lower than several CNN-based approaches; limited disease categories

14.	K. Sathya et al., IEEE Access, 2025	Attention-Augmented Residual Network + cGAN augmentation + Faster R-CNN Dataset: Augmented plant-disease image dataset using GAN-generated samples	98.78% classification accuracy Faster-RCNN effectiveness improved by 23.84%	Strong feature focus; augmentation improves generalization	Increased training/inference complexity
15.	B. Ramana Reddy et al., IEEE Access, 2025	Lightweight custom CNN + OpenCV severity estimation + mobile application Dataset: PlantVillage dataset	92.06% test accuracy 90% precision/recall F1	Practical mobile deployment; provides disease severity estimation	Binary healthy/diseased classification rather than detailed disease-class recommendation
16.	A. Khaliq et al., IEEE Access, 2025	AI + IoT + XAI; TabNet for fertilizer recommendation, SwiFT for crop recommendation and TabNet soil-health analysis Dataset: Temperature, humidity, moisture, soil type, crop type, N/P/K and fertilizer data	TabNet fertilizer recommendation 99.3% accuracy; crop recommendation 98.75%	Directly combines crop and fertilizer recommendations with XAI	Dataset/domain dependency; fertilizer labels are limited compared with real-world agrochemical choices
17.	J. Madhuri et al., IEEE Access, 2025	Improved Deep Belief Network using Gaussian RBM + Ranger Optimizer for crop recommendation Dataset: Soil, weather and crop datasets	Proposed IDBN outperformed conventional DBN and comparison models	Uses physical + chemical soil and climate features	Primarily crop recommendation rather than agrochemical recommendation

18	A. Badshah et al., IEEE Access, 2024	ETC, LR, DT, RF, KNN, GNB and SVM with feature engineering + XAI for crop recommendation; SVR for yield prediction Dataset: Kaggle crop recommendation dataset	RF:99.7% accuracy for crop recommendation; SVR: 99.9% R ² -based performance for wheat yield	Strong ML comparison + XAI; useful for agricultural decision support	Recommendation is crop-focused rather than direct chemical recommendation
19	WB-CPI research group, IEEE Access, 2021	MapReduce + K-means + recommender system using weather, soil, seed and crop-production information Dataset: Agricultural data	Clustering/recommendation evaluation; no single universal accuracy emphasized	Combines large-scale agricultural data with recommendation	Regional data; crop recommendation rather than agrochemical recommendation
20	U.umar T.A. Sardjono, H. Kusuma, IEEE Access, 2024	Ontology + environmental sensors + YOLOv8 + semantic recommendation framework for melon cultivation Dataset: environmental sensor data	System evaluated for recommendation accuracy/reliability and practical farming efficiency	Context-aware recommendations for seed selection, soil, irrigation and pest control	Focused on melon cultivation; ontology requires domain-specific knowledge

Research Gaps

- **Disease detection and treatment recommendation are mostly separate**
 - Existing systems usually detect crop diseases or recommend fertilizers separately.
 - Very few systems combine both functions into one system.
- **Disease detection often stops at identifying the disease**
 - Many AI models provide only the disease name.
 - They do not provide suitable treatment or agrochemical information after detection.
- **Recommendation systems do not usually use disease information**
 - Fertilizer recommendation systems mainly use N, P, K, pH, soil, weather and crop data.
 - Disease information from leaf images is generally not considered.
- **Dependence on controlled datasets**
 - Many disease-detection models are trained using datasets such as PlantVillage.
 - These datasets may not fully represent real farm conditions.
- **Difficulty in real-field conditions**
 - Changes in lighting, background, leaf orientation, image quality and disease severity can reduce model performance.
- **Limited crop and regional applicability**
 - Some existing systems are developed only for specific crops or geographical regions.
 - This limits their use in different agricultural environments.
- **Disease severity is not consistently considered**
 - Some systems identify the disease but do not consider how severe the disease is when providing treatment information.
- **Limited explainability**
 - Some AI models give predictions without clearly explaining the reason behind the prediction or recommendation.
- **High computational requirements**
 - Advanced CNN, Transformer and ensemble models may require high computational resources.

- This can make deployment on low-resource devices difficult.
- **Need for reliable agricultural knowledge**
- Agrochemical recommendations need accurate and crop-specific agricultural information.
- Incorrect recommendations can reduce the usefulness and reliability of the system.

Overall Research Gap

Existing research shows strong results in crop disease detection and agricultural recommendation, but these functions are mostly developed independently. There is a need for a unified system that can detect crop diseases from leaf images and provide suitable agrochemical treatment information based on the detected disease. The proposed system addresses this gap by integrating AI-based disease detection with smart agrochemical recommendation in a single farmer-oriented decision-support system.

Base Paper 1

Plant Disease Identification Using a Novel Convolutional Neural Network

Link: <https://ieeexplore.ieee.org/document/9674894>

Authors: Sk Mahmudul Hassan, Arnab Kumar Maji

Journal: IEEE Access

Year: 2022

Method: CNN with Inception, residual connections and depthwise-separable convolution

Datasets: PlantVillage, rice disease and cassava disease datasets

Best Accuracy:

- PlantVillage: **99.39%**
- Rice: **99.66%**
- Cassava: **76.59%**

Used as base for: AI-based crop disease detection

Why we selected: Provides a strong deep-learning foundation for identifying diseases from crop leaf images.

Base Paper 2

A Recommendation Model of Rice Fertilization Using Knowledge Graph and Case-Based Reasoning

Link: <https://doi.org/10.1016/j.compag.2024.108751>

Authors: Weixi Ge, Jun Zhou, Pengyuan Zheng, Licun Yuan, Luke Toroitich Rottok

Journal: Computers and Electronics in Agriculture

Publisher: Elsevier

Year: 2024

Method: Knowledge Graph + Case-Based Reasoning

Data: Rice fertilization cases and agricultural/fertilization knowledge

Purpose: Generates suitable rice fertilization plans.

Used as base for: Smart agrochemical/fertilizer recommendation

Why we selected: Shows how agricultural knowledge can be used to generate intelligent and context-based recommendations.